FUTURE HEALTH | A GLOBAL INITIATIVE BY ABU DHABI | H2S HACK2SKILL

Future Health Digital Innovation Challenge

From signals to solutions.

Co-creator Microsoft

Co-creator inception42

Co-creator Mohamed bin Zayed University of Artificial Intelligence

Team Name: **Riaya (رعاية)**
Team Size: 1 (solo entry: Amritha S)
Problem Statement: Identifying male reproductive health risks early: a private, culturally sensitive front door that screens (never diagnoses), explains every signal, and routes men to the right level of care with a licensed doctor approving every specialist referral.

Team members*
[Name, country, role, organization, age confirmation]

S.No	Name	Country	Role	Organization	Age confirmation
1	Amritha S	**[ADD COUNTRY]**	Founder; product, research and full-stack engineering	**[ADD ORGANISATION]**	**[CONFIRM 18+]**

Solo entry. Every part of Riaya (research, evidence map, design, code, tests and documentation) was produced by Amritha S.

Team Capability Summary*

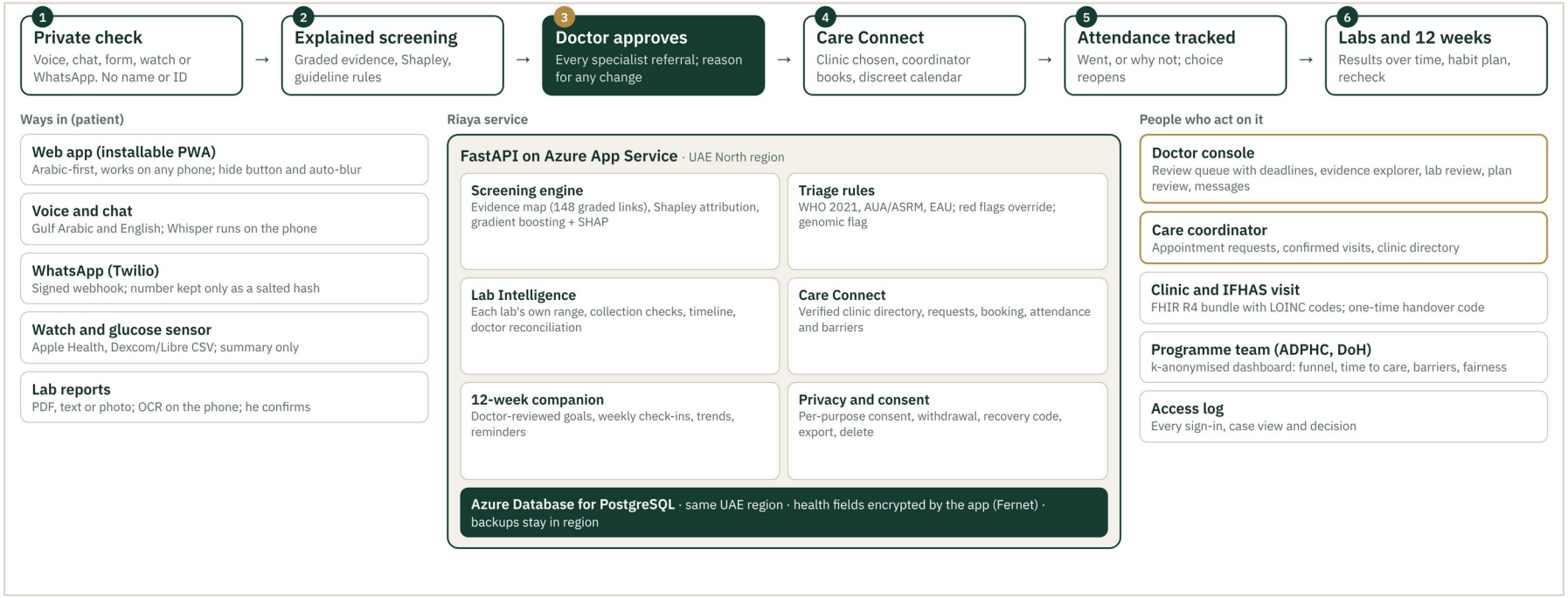
Maximum 1,000 characters

Solo builder covering health research, product design and software engineering. For Riaya I reviewed and graded 95 published sources (WHO 2021, AUA/ASRM, EAU) into an evidence map of 148 cited links; designed an Arabic-first, privacy-first patient experience; and built the full stack: Python/FastAPI services, PostgreSQL with encryption at rest, explainable machine learning (scikit-learn, SHAP, Shapley attribution), guideline-based triage with human sign-off, lab-report extraction, FHIR R4 hand-off, WhatsApp integration and a clinician console. Quality practice: 32 automated end-to-end tests run on SQLite and PostgreSQL in GitHub Actions, browser testing with Playwright, a security review (lockouts, session limits, upload limits, CSP) and written operations, privacy and Azure UAE deployment guides.

Shown in the working prototype

- **Research:** 95 sources, graded A–D
- **Clinical logic:** guideline triage, red flags
- **Data science:** explainable models, validation
- **Engineering:** FastAPI, PostgreSQL, PWA
- **Privacy and security:** consent, encryption, audit
- **Design:** Arabic-first, accessible
- **Delivery:** CI, tests, Azure UAE template

Architecture or workflow diagram*



Deployment: azure/main.bicep provisions App Service and PostgreSQL in Azure UAE North or UAE Central. The public prototype runs on test data only and says so on every page.

Target users and intended outcome*

Primary users

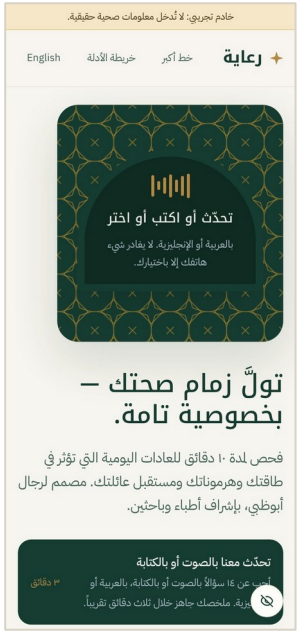
- **Men in Abu Dhabi,** especially 20–45, Emirati and resident, Arabic or English speaking, who avoid clinics for reasons of privacy or embarrassment
- **Couples planning a family,** who can share a status-only partner link

Professional users

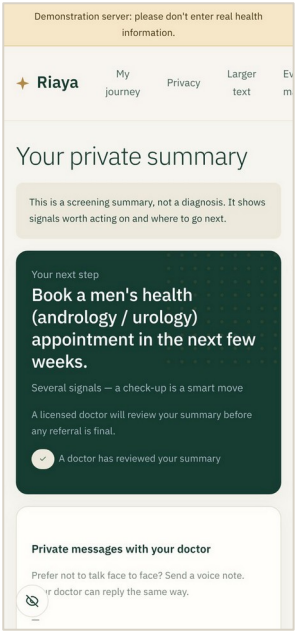
- **Doctors** (family medicine, urology/andrology) who review referrals, labs and plans
- **Care coordinators** who book and track clinic visits
- **ADPHC / DoH programme teams** using the anonymised dashboard; IFHAS providers receiving the FHIR hand-off

Intended outcome

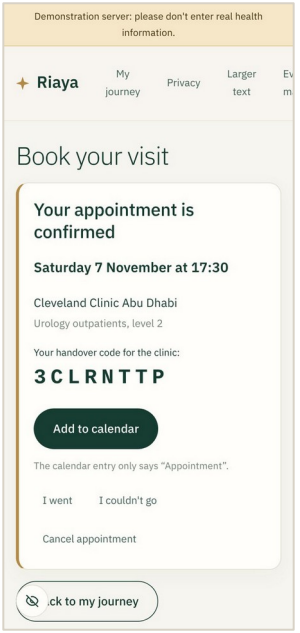
Men identified earlier and sent to the right level of care, with a doctor approving every specialist referral, fewer lost between referral and visit, and habits changed over the 12 weeks sperm take to develop.



Arabic-first, private



Explained summary



Visit booked

How will success be measured?*

Area	Measure (computed live in the clinician metrics and programme dashboard)	Proposed pilot target
Reach	Checks started and completed; median minutes to complete; share who used voice, WhatsApp or watch	≥ 70% completion, ≤ 10 min
Safety	Under-triage on blinded vignettes reviewed by UAE clinicians; red flags missed	0 red flags missed
Clinical agreement	Share of suggested referrals approved without change; reasons for overrides	≥ 80% approved unchanged
Time to care	Median hours to doctor review; median days from check to clinic visit	Review ≤ 48 h; visit ≤ 21 days
Closing the loop	Care Connect booking rate, attendance rate, and why visits did not happen	≥ 60% of referrals attended
Behaviour change	12-week check-ins kept; change in cigarettes, sleep and weight; 3-month recheck rate	≥ 50% log 8+ weeks
Lab quality	Share of extracted lab values doctors accept unchanged	≥ 95% accepted
Trust and fairness	In-app trust, clarity, comfort (1–5); referral rates by age band and background	Trust ≥ 4 / 5; no unexplained gaps

Targets are proposals to agree with DoH Abu Dhabi and ADPHC before the pilot. Every measure above is already calculated by the prototype.

Evidence supporting the approach*

Citations, studies, benchmarks, or prior testing.

Published evidence behind the rules and the map

- **WHO (2021)** Laboratory manual for human semen, 6th ed.: semen reference limits and collection (2–7 days abstinence)
- **Schlegel et al. (2021)** AUA/ASRM male infertility guideline; EAU (2024) sexual and reproductive health guideline: evaluation pathway, repeat testing
- **Mulhall et al. (2018) AUA; Bhasin et al. (2018) Endocrine Society:** testosterone confirmed by two early-morning tests
- **Sharma et al. (2016) Eur Urol:** smoking and semen quality, 20 studies, 5,865 men
- **Sermondade et al. (2013) HRU:** obesity and lower sperm count
- **Mínguez-Alarcón et al. (2018) Hum Reprod:** tight underwear and testicular function
- **Levine et al. (2023) HRU:** sperm counts declining worldwide; WHO (2023): ~1 in 6 people affected by infertility

All 95 sources, each link graded A–D, are listed in docs/REFERENCES.md.

Prior testing (honest limits)

- **26 / 26** guideline scenarios gave the expected next step, with 0 under-triage. Scenarios were written by the team, so this checks the rules, not accuracy
- **AUC 0.62–0.69** on real data (UCI Fertility, n = 100, small)
- **AUC 0.68** on a simulated cohort of 20,000 men using literature effect sizes, with calibration and subgroup fairness tables
- **AUC 0.66** from the evidence map alone, untrained: the map agrees with the literature
- **32 automated tests** on SQLite and PostgreSQL; end-to-end browser checks

Next: blinded vignette review by UAE clinicians, then pilot data.

Data sources and legal right to use them*

List source, type, owner, sensitivity, permissions, and whether personal data is used.

Source	Type	Owner	Sensitivity	Permission and handling	Personal?
Answers, voice and chat	Self-reported health	The man	High (health)	Consent per purpose; no name, phone or ID; encrypted; withdraw, export, delete	Yes, pseudonymous
Watch and glucose exports	Device data	The man	High	Uploaded only with device-data consent; summary kept, file discarded	Yes
Lab reports	Clinical results	The man / his lab	High	He confirms each value; file never stored; doctor reconciles	Yes
WhatsApp messages	Conversation	The man (Twilio processes)	High	Number kept only as a salted hash; "delete" erases all; approved channel needed for pilot	Yes
Published literature	95 papers and guidelines	Publishers	None	Cited, not reproduced	No
UCI Fertility dataset	Open dataset, n = 100	UCI ML Repository	Anonymised	CC BY 4.0, attributed (Gil et al. 2012)	No
Simulated cohort	Synthetic, 20,000	Created by the team	None	Generated from literature effect sizes	No
Clinic directory	Public facts	Each provider	Low	Name, area, sector, website only; no invented phone numbers or availability	No

Risks and safeguards*

Privacy; security; bias; explainability; misuse; human oversight; failure modes.

Risk	Safeguard built into the prototype
Privacy	No name, phone or ID. Consent per purpose, withdrawable. Encryption at rest. One-time share codes. Hide button and auto-blur. Status-only partner link. Export and delete everything
Security	Doctor lockout after 5 failures; 8 h sessions, 30 min idle; strong-password rule; access log of every view and decision; upload and zip-bomb limits; CSP and HSTS
Bias	Results by age band and background in the fairness tables; k-anonymised dashboard (groups under 5 hidden); real-data limits stated
Explainability	Every finding shows the evidence, its grade, sources, and limits; Shapley attribution; unknown data shown as unknown, never as normal
Misuse	Screening only; no fertility prediction score; calendar entries say only "Appointment"; partner link shows no answers or results
Human oversight	A doctor approves every specialist referral, lab correction and 12-week plan; any change needs a written reason
Failure modes	Red flags skip scoring and go straight to urgent or emergency advice; implausible lab values are refused; a single abnormal result prompts a repeat test, not a conclusion; video stays off unless an approved UAE service is set up

Hardware requirements

Required if wearable, sensor, connected device, or other hardware is there. Include sourcing and safety.

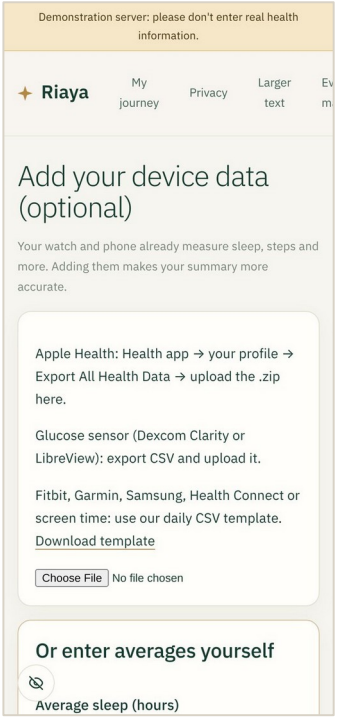
No hardware is required or supplied.

Riaya is software only: a web app that installs on any phone without an app store, plus WhatsApp.

Optional data from devices men already own

- **Apple Watch / iPhone:** Health app export (.zip)
- **Glucose sensors:** Dexcom Clarity or LibreView CSV export
- **Fitbit, Garmin, Samsung:** daily CSV template
- **Home sperm tests:** results typed in, or a lab report uploaded

Riaya never connects to or controls a device. It reads files the man chooses to export, so there is no sourcing, device safety or regulatory burden on hardware.



Path to pilot and scale*

Stakeholders, integration, regulatory needs, timeline assumptions, and sustainability.

Months 0–1

Ready to pilot

Azure UAE deployment, ADHICS review, DPIA, SaMD classification confirmed with the regulator, blinded vignette review by UAE clinicians

Months 1–6

IFHAS pilot

Optional pre-screen at 2–3 IFHAS providers; FHIR hand-off into the visit; measure completion, time to care, agreement, trust

Months 6–12

Specialist pathway

Direct booking with SEHA / M42 urology and andrology clinics; outcomes fed back; retrain models on real paired data

Year 2+

Scale

Emirate-wide through IFHAS and Malaffi; UAE PASS for clinicians; Urdu, Hindi and Malayalam

- **Stakeholders:** DoH Abu Dhabi, ADPHC (IFHAS), SEHA and M42 clinics, Malaffi, Thiqa programme
- **Integration:** FHIR R4 bundle with LOINC codes; one-time handover codes; Azure UAE hosting
- **Regulation:** UAE Federal Law No. 2 of 2019 (health data in the UAE); DoH ADHICS; DoH AI policy; SaMD status to confirm
- **Sustainability:** funded per completed screen by the programme or as a licence to IFHAS providers; low fixed cost (one small App Service and database)
- **Assumptions:** programme sponsor in place; clinics accept referrals; hosting approval within the first month
- **Why it scales:** no hardware, no app store, works on WhatsApp, and the clinical rules live in one versioned, cited file

Pre-existing IP and third-party materials*

Identify ownership, licenses, open-source components, and restrictions.

Ownership

All code, the evidence map, content, design and documentation were created by Amritha S, who owns them. No employer or third party has a claim.

Restrictions

- **No proprietary third-party code** is included
- **psycopg (LGPL-3.0)** is used unmodified as a library
- **Fonts (OFL)** are shipped with their licence
- **Twilio** is a paid service under its own terms
- **Guidelines and papers** are cited, not reproduced

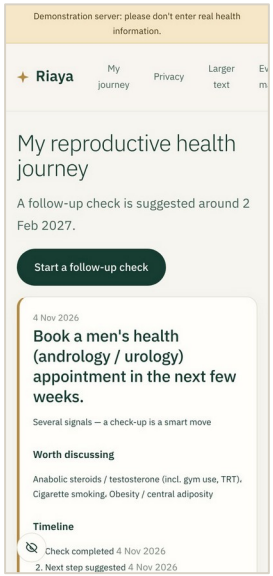
Component	Use	Licence
FastAPI, Pydantic, Uvicorn	Web service	MIT / BSD-3
scikit-learn, NumPy, pandas, joblib	Models and data	BSD-3
SHAP	Model explanations	MIT
cryptography	Encryption at rest	Apache-2.0 / BSD
psycopg 3	PostgreSQL driver	LGPL-3.0
pypdf, httpx	PDF reading, HTTP	BSD-3
tesseract.js	OCR on the phone	Apache-2.0
transformers.js + Whisper	Speech on the phone	Apache-2.0 / MIT
IBM Plex, Noto Kufi Arabic	Fonts	SIL OFL 1.1
qrcodejs	QR codes	MIT
UCI Fertility dataset	Model training	CC BY 4.0

Conflict of interest declaration*

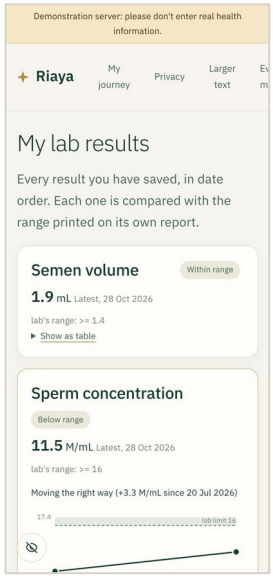
Disclose relationships with organizers, partners, or judges.

None. No member of our team has a conflict of interest to declare.

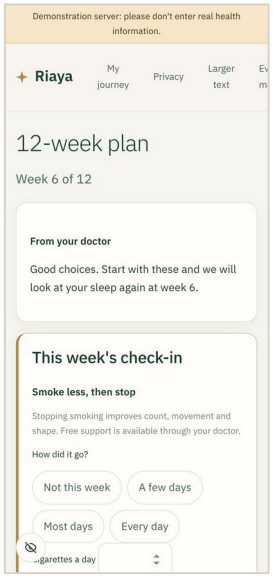
Amritha S is not employed by or contracted to the Department of Health – Abu Dhabi, Future Health, Microsoft, Inception42, MBZUAI or Hack2skill, has no family or close personal relationship with a judge, organiser or collaborator, holds no financial interest in an organisation running or judging the challenge, and has had no access to non-public information about the challenge or its evaluation.



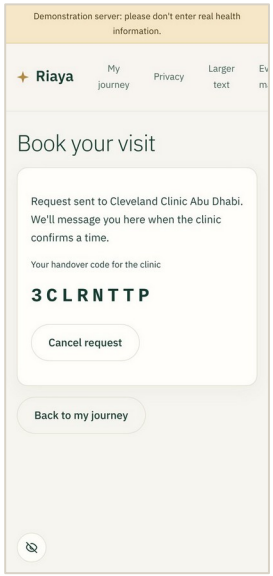
Journey



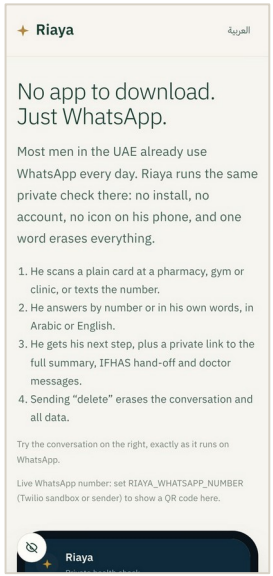
Lab results over time



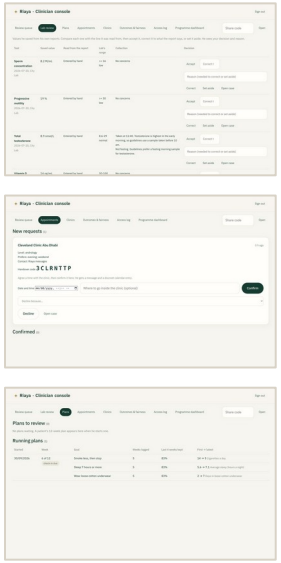
12-week check-in



Clinic request sent



WhatsApp



Doctor and coordinator

Additional slide: what is built and working today

Code: github.com/Amritha902/health_innov-.- **Prototype:** **[ADD PROTOTYPE LINK]**

FUTURE HEALTH | A GLOBAL INITIATIVE BY ABU DHABI | H2S HACK2SKILL

Future Health Digital Innovation Challenge

From signals to solutions.

Thank you

Co-creator

Microsoft

Co-creator

inception42

Co-creator

Mohamed bin Zayed University of Artificial Intelligence